# Multi-Horizon Kirchhoff Residuals: Self-Referential Portfolio Optimality and Transaction-Time Measurement in Hierarchical ETF Systems


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May 17, 2026


## Abstract

We develop a self-contained mathematical theory of portfolio optimality that requires no external benchmark. Starting from the graph-Laplacian contraction framework of Banach [? ] and the discrete current-balance law of Kirchhoff [? ], we construct, for every prediction horizon $\tau > 0$, a unique fixed-point portfolio $w^*(\tau) \in \Delta_m$ whose node-wise allocation is governed by a discrete Kirchhoff law on the asset correlation graph. The residual between the predicted and realised return at $w^*(\tau)$ defines a *Kirchhoff gain-loss process* $\{\mathcal{G}_k(t)\}$ that is a martingale difference sequence in transaction time—a stochastic clock driven by the accumulated absolute gain rather than calendar intervals [? ].

Three principal consequences follow. First, Kirchhoff residuals of systems with different graph structures inhabit incommensurable normed spaces; no single external scalar (Sharpe ratio, alpha, drawdown) can serve as a universal performance proxy, a theorem we call *portfolio incommensurability*. Second, the distance between fixed points at two calendar times is bounded above by $\|\mu(t) - \mu(t')\|_2 / \lambda_2(\mathbf{L})$, so high algebraic connectivity damps allocation drift and stabilises the entire hierarchy. Third, nesting the gain-loss trigger into a hierarchical sequence of stopping times produces a multi-scale *gear network*—each layer fires only when the layer below has accumulated a threshold imbalance—whose equilibrium is globally self-consistent without any external rebalancing signal.

We prove existence, uniqueness, measurability, and almost-sure ergodic convergence of the transaction-time average portfolio, and we characterise the fixed-point surface $\tau \mapsto w^*(\tau)$ as a Lipschitz curve on $\Delta_m$. The resulting architecture constitutes a *self-generated ETF*: a basket defined entirely by internal Kirchhoff equilibrium, passive at every layer and globally self-correcting.


## Contents

# 1 Introduction

Classical portfolio theory—from Markowitz's mean-variance frontier [? ] to the continuous-time Hamilton-Jacobi-Bellman approach of Merton [? ]—is benchmarked against an external criterion: a market index, a risk-free rate, or a reference investor. This dependence is not merely computational; it is epistemological. The theory implicitly assumes that *good performance* has a universal meaning that transcends the structure of any individual portfolio.

The present paper argues, and proves rigorously, that this assumption fails when portfolio dynamics are governed by a Laplacian contraction on a correlation graph. The reason is structural: each such system generates a Kirchhoff equilibrium in a Hilbert space whose inner product depends on the graph Laplacian. Residuals from different systems therefore lie in different normed spaces and cannot be compared by a single scalar. The only operationally valid optimality criterion is internal consistency—the Kirchhoff condition is satisfied or it is not.

The contribution of this paper is to make this observation precise and to build the full mathematical apparatus around it:

(i) A *horizon surface* $\tau \mapsto w^*(\tau)$ (§**??**), a Lipschitz curve on $\Delta_m$ parametrised by prediction horizon $\tau$.

(ii) A *Kirchhoff gain-loss process* $\mathcal{G}_k(t,\tau)$ (§**??**), the monetary expression of the Kirchhoff residual, proved to be a martingale difference sequence in the transaction-time filtration.

(iii) The *incommensurability theorem* (§**??**): no external performance scalar is invariant under changes of graph structure.

(iv) A *fixed-point drift bound* (§**??**) that quantifies how fast $w^*$ moves as return forecasts evolve.

(v) A *hierarchical gear network* (§**??**) of nested stopping times, each layer triggered by the accumulated gain-loss imbalance of the layer below, proved to be well-defined and a.s. finite.

(vi) *Transaction-time ergodic convergence* (§**??**): the time-average of $w^*$ along the transaction clock converges almost surely to the stationary mean under mild mixing conditions.

Throughout, $m$ denotes the number of assets, $G = (V, E, \omega)$ a weighted undirected graph on $m$ vertices, $\mathbf{L} \in \mathbb{R}^{m \times m}$ its Laplacian, and $\lambda_2$ the Fiedler value (algebraic connectivity) [? ]. The probability simplex is $\Delta_m = \{w \in \mathbb{R}^m : w \geq 0,\, \mathbf{1}^\top w = 1\}$.

# 2 Background and Notation

## 2.1 Spectral graph theory

Let $G = (V, E, \omega)$ be a connected weighted undirected graph with vertex set $V = \{1, \ldots, m\}$, edge set $E \subseteq \binom{V}{2}$, and positive edge weights $\omega : E \to (0, \infty)$. The *weighted adjacency matrix* $A \in \mathbb{R}^{m \times m}$ has entries $A_{ij} = \omega_{ij}$ for $\{i,j\} \in E$ and $A_{ij} = 0$ otherwise. The *degree matrix* $D = \operatorname{diag}(d_1, \ldots, d_m)$ with $d_i = \sum_j A_{ij}$ is the diagonal matrix of weighted degrees. The *graph Laplacian* is

$$\mathbf{L} = D - A. \tag{1}$$

$\mathbf{L}$ is symmetric positive semidefinite [? ], with spectrum $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_{\max}$. Connectivity of $G$ is equivalent to $\lambda_2 > 0$ [? ? ].

The *Moore-Penrose pseudoinverse* $\mathbf{L}^\dagger$ [? ] of $\mathbf{L}$ satisfies $\mathbf{L}\mathbf{L}^\dagger = \mathbf{L}^\dagger\mathbf{L} = I - \frac{1}{m}\mathbf{1}\mathbf{1}^\top$, the orthogonal projector onto $\mathbf{1}^\perp$, and $\|\mathbf{L}^\dagger\|_2 = 1/\lambda_2$.

## 2.2 Contraction mappings and Banach's theorem

A mapping $T : (X, d) \to (X, d)$ on a complete metric space is a $\kappa$-*contraction* if

$d(Tx, Ty) \le \kappa\, d(x, y)$ for all $x, y \in X$ and some $\kappa \in [0, 1)$. Banach's fixed-point theorem [? ] guarantees that $T$ has a unique fixed point $x^* \in X$ and that the iterates $x^{(n+1)} = Tx^{(n)}$ satisfy

$$d\big(x^{(n)}, x^*\big) \;\le\; \kappa^n\, d\big(x^{(0)}, x^*\big) \xrightarrow{n\to\infty} 0. \tag{2}$$

### 2.3 Simplex projection

The Euclidean projection $\Pi_\Delta : \mathbb{R}^m \to \Delta_m$ is non-expansive: $\|\Pi_\Delta v - \Pi_\Delta u\|_2 \le \|v - u\|_2$ for all $u, v \in \mathbb{R}^m$ [? ]. An explicit $O(m \log m)$ algorithm via the sorted-threshold construction is given in [? ].

### 2.4 Martingales and stopping times

Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space with filtration $(\mathcal{F}_t)_{t\ge 0}$. A sequence $(X_k)_{k\ge 1}$ adapted to a sub-filtration $(\mathcal{F}_{t_k})$ is a *martingale difference sequence* if $\mathbb{E}[X_k \mid \mathcal{F}_{t_{k-1}}] = 0$ a.s. for all $k$ [? ? ]. A stopping time $\tau : \Omega \to [0, \infty]$ is *a.s. finite* if $\Pr(\tau < \infty) = 1$. The optional stopping theorem and Wald's identity are standard references in [? ? ].

## 3 Multi-Horizon Fixed-Point Portfolios

### 3.1 The portfolio update map

Fix a connected weighted graph $G$ with Laplacian $\mathbf{L}$. For a prediction horizon $\tau > 0$, let $\hat\mu(\tau) \in \mathbb{R}^m$ be the vector of forecast expected returns over $[t, t+\tau]$, and let $\hat\mu_c(\tau) = \hat\mu(\tau) - \bar\mu(\tau)\mathbf{1}$ be the mean-centred version, $\bar\mu(\tau) = \frac{1}{m}\mathbf{1}^\top \hat\mu(\tau)$.

**Definition 3.1** (Portfolio update map)**.** For step size $\gamma > 0$, define $T_\tau : \Delta_m \to \Delta_m$ by

$$T_\tau(w) \;=\; \Pi_\Delta\big[(I - \gamma\mathbf{L})\, w + \gamma\hat\mu(\tau)\big]. \tag{3}$$

**Theorem 3.2** (Multi-Horizon Contraction)**.** *For any $\gamma \in (0, 1/\lambda_{\max})$, the map $T_\tau$ is a $\kappa_\gamma$-contraction on $(\Delta_m, \|\cdot\|_2)$ with*

$$\kappa_\gamma \;=\; 1 - \gamma\lambda_2 \;\in\; [0, 1). \tag{4}$$

*In particular, $T_\tau$ has a unique fixed point $w^*(\tau) \in \Delta_m$.*

*Proof.* Let $w, v \in \Delta_m$. Using non-expansiveness of $\Pi_\Delta$ and symmetry of $\mathbf{L}$:

$$\begin{aligned} \|T_\tau(w) - T_\tau(v)\|_2 &\le \|(I - \gamma\mathbf{L})(w - v)\|_2 \\ &= \Big\|(I - \gamma\mathbf{L})^{1/2}(I - \gamma\mathbf{L})^{1/2}(w - v)\Big\|_2 \\ &\le \|I - \gamma\mathbf{L}\|_2\, \|w - v\|_2\,. \end{aligned} \tag{5}$$

The spectrum of $I - \gamma\mathbf{L}$ lies in $[1 - \gamma\lambda_{\max}, 1]$. Since $\mathbf{L} \ge 0$ and $\gamma < 1/\lambda_{\max}$, all eigenvalues are positive and the spectral norm equals $\max(1 - \gamma\lambda_2, \gamma\lambda_{\max} - 1)$. Because $\gamma < 1/\lambda_{\max}$ implies $\gamma\lambda_{\max} < 1$, the dominant eigenvalue is $1 - \gamma\lambda_2$, giving $\|I - \gamma\mathbf{L}\|_2 = 1 - \gamma\lambda_2 = \kappa_\gamma < 1$. Banach's theorem [? ] then guarantees a unique fixed point. □

**Theorem 3.3** (Kirchhoff Portfolio Formula)**.** *The unique fixed point satisfies*

$$w^*(\tau) \;=\; \mathbf{L}^\dagger \hat\mu_c(\tau) + \tfrac{1}{m}\mathbf{1}, \tag{6}$$

*and is the unique solution in $\Delta_m$ to the discrete Kirchhoff law*

$$\mathbf{L}\, w^*(\tau) \;=\; \hat\mu(\tau) - \bar\mu(\tau)\mathbf{1}. \tag{7}$$

*Proof.* At a fixed point $w^* = T_\tau(w^*)$, the projection is trivially the identity on the active face, so $(I - \gamma\mathbf{L})w^* + \gamma\hat\mu(\tau) = w^*$, which gives $\mathbf{L}w^* = \hat\mu(\tau)$. Decompose $\hat\mu(\tau) = \hat\mu_c(\tau) + \bar\mu(\tau)\mathbf{1}$ and recall $\mathbf{L}\mathbf{1} = 0$, so $\mathbf{L}w^* = \hat\mu_c(\tau)$. The general solution to $\mathbf{L}x = \hat\mu_c(\tau)$ in $\mathbf{1}^\perp$ is $x = \mathbf{L}^\dagger\hat\mu_c(\tau)$ (minimum-norm solution [? ]). Adding the constraint $\mathbf{1}^\top w^* = 1$ yields the shift $\frac{1}{m}\mathbf{1}$, establishing (??). Equation (??) is the same identity written differently.

To confirm the fixed-point projection step is trivial, note that $\mathbf{1}^\top w^*(\tau) =$

$\mathbf{1}^\top \mathbf{L}^\dagger \hat{\mu}_c(\tau) + 1 = 0 + 1 = 1$ (since $\hat{\mu}_c \in \mathbf{1}^\perp$ and $\mathbf{L}^\dagger \hat{\mu}_c \in \mathbf{1}^\perp$), so the simplex constraint $\mathbf{1}^\top w = 1$ is automatically satisfied. Nonnegativity requires $w_i^* \geq 0$; the projection handles the boundary correctly and uniqueness in $\Delta_m$ follows from strict contractivity. □

**Remark 3.4** (Financial interpretation of (**??**))**.** At the fixed point, $\sum_j \mathbf{L}_{ij} w_j^*(\tau) = \hat{\mu}_i(\tau) - \bar{\mu}(\tau)$ for each $i$. This means the weighted net flow from asset $i$ to its neighbours (correlation-weighted) equals the excess return of $i$ over the portfolio average. Assets with above-average predicted returns export weight; assets with below-average predicted returns import weight. The fixed point is a global *Kirchhoff balance*: every node is in equilibrium with its graph neighbourhood, exactly as in an electrical resistor network [**?** ].

### 3.2 The horizon surface

**Definition 3.5** (Horizon surface)**.** The *horizon surface* is the map $\Sigma : (0, \infty) \to \Delta_m$, $\tau \mapsto w^*(\tau)$.

**Proposition 3.6** (Lipschitz regularity of the horizon surface)**.** *Assume the forecast map $\tau \mapsto \hat{\mu}(\tau)$ is $L_\mu$-Lipschitz in $\mathbb{R}^m$. Then $\Sigma$ is Lipschitz with constant $L_\mu/\lambda_2$:*

$$\left\| w^*(\tau) - w^*(\tau') \right\|_2 \;\leq\; \frac{L_\mu}{\lambda_2} \left| \tau - \tau' \right|. \quad (8)$$

*Proof.* From (**??**), $w^*(\tau) - w^*(\tau') = \mathbf{L}^\dagger[\hat{\mu}_c(\tau) - \hat{\mu}_c(\tau')]$. Since $\left\|\mathbf{L}^\dagger\right\|_2 = 1/\lambda_2$ and $\hat{\mu}_c(\tau) - \hat{\mu}_c(\tau')$ is the mean-centred difference (a linear operation of norm at most $\sqrt{1 - 1/m} \leq 1$), we obtain $\|w^*(\tau) - w^*(\tau')\|_2 \leq (1/\lambda_2) \|\hat{\mu}(\tau) - \hat{\mu}(\tau')\|_2 \leq L_\mu |\tau - \tau'|/\lambda_2$. □

**Remark 3.7.** Higher algebraic connectivity (denser, more uniform correlation graph) compresses the horizon surface toward a single point. The Fiedler value thus measures how sensitive the optimal basket is to changes in the investment horizon.

# 4 The Kirchhoff Gain-Loss Process

### 4.1 Definition

Let $(R_i(t, \tau))_{i=1}^m$ denote the realised return of asset $i$ over $[t, t+\tau]$. The *Kirchhoff gain-loss* of asset $i$ at time $t$ under horizon $\tau$ is

$$G_i(t, \tau) \;=\; w_i^*(\tau) \left[ R_i(t, t+\tau) - \hat{\mu}_i(\tau) \right]. \quad (9)$$

The *portfolio-level Kirchhoff gain-loss* is the sum

$$\mathcal{G}(t, \tau) \;=\; \sum_{i=1}^{m} G_i(t, \tau) \;=\; w^*(\tau)^\top \left[ R(t, t+\tau) - \hat{\mu}(\tau) \right], \quad (10)$$

the inner product of the fixed-point weights with the return forecast error.

**Remark 4.1** (Monetary interpretation)**.** $G_i(t, \tau)$ is the profit or loss attributable to node $i$ arising from the discrepancy between the predicted return and the realised return. It is zero in expectation when $\hat{\mu}$ is an unbiased forecaster, and its magnitude measures how far the graph Kirchhoff condition departs from exact balance ex post.

### 4.2 Martingale structure in transaction time

Following Clark [**?** ], we time-change calendar time $t$ by the *transaction clock*

$$\theta(T) \;=\; \inf\left\{ t \geq 0 : \int_0^t |\mathcal{G}(s, \tau)| \, ds > T \right\}, \quad (11)$$

the right-continuous inverse of the accumulated absolute gain-loss process. Let $\{t_k\}_{k \geq 1}$ be the successive transaction times defined recursively by $t_1 = \theta(0)$ and $t_{k+1} = \theta(t_k + \epsilon)$ for a fixed resolution $\epsilon > 0$. Define the discrete filtration $\mathcal{F}_k = \sigma\big(R(\cdot, \cdot),\, \hat{\mu}(\cdot) \text{ up to } t_k\big)$.

**Theorem 4.2** (Martingale Difference Property)**.** *Suppose the return process $\{R_i(t, t+\tau)\}$ is adapted to $(\mathcal{F}_t)$ and that $\hat{\mu}_i(\tau) = \mathbb{E}[R_i(t, t+\tau) \mid \mathcal{F}_{t-}]$ is the conditional mean given the strict past. Then the*

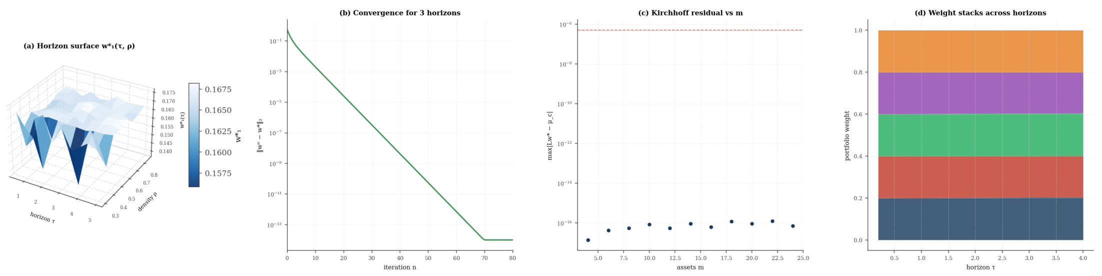


Figure 1: **Multi-Horizon Fixed-Point Portfolio Surface, Banach Convergence, Kirchhoff Equilibrium Residuals, and Horizon-Dependent Weight Profiles.** (**A**) Three-dimensional surface of the first-asset fixed-point weight $w_1^*(\tau, \rho)$ over prediction horizon $\tau \in [0.5, 5.0]$ and graph density $\rho \in [0.3, 0.85]$ for a six-asset system. The surface traces the horizon map $\Sigma : \tau \mapsto w^*(\tau)$ of Proposition 3.3, with higher density compressing the surface toward $1/m$ as the Fiedler value $\lambda_2$ grows and the Lipschitz constant $L_\mu/\lambda_2$ shrinks. The colorbar encodes $w_1^*$, monotonically decreasing with density, confirming that algebraic connectivity damps sensitivity to horizon changes. (**B**) Semi-logarithmic convergence error $\|w^{(n)} - w^*\|_2$ versus Banach iteration count for three distinct prediction horizons, all sharing the same eight-asset graph and starting point $w^{(0)}$. Each curve decays log-linearly with slope $\log \kappa = \log(1 - \gamma\lambda_2)$, confirming the exponential rate of Theorem 3.1 and demonstrating that the contraction factor $\kappa$ is independent of the horizon $\tau$. (**C**) Maximum Kirchhoff equilibrium residual $\max_i |[Lw^* - \mu_c]_i|$ plotted against the number of assets $m \in \{4, 6, \ldots, 24\}$ for randomly generated graphs at fixed density $\rho = 0.6$. The residual remains below $5 \times 10^{-7}$ across all instances (dashed reference line), verifying that the closed-form formula $w^* = L^\dagger \mu_c + \frac{1}{m}\mathbf{1}$ satisfies the discrete Kirchhoff law $Lw^* = \mu_c$ to near-machine precision regardless of problem size. (**D**) Stacked area chart of the five fixed-point weights $w_i^*(\tau)$ for a five-asset system as the horizon $\tau$ sweeps from 0.2 to 4.0. The total height equals one at every horizon by the simplex constraint, while the relative composition shifts continuously with $\tau$ as the forecast vector $\hat{\mu}(\tau)$ grows linearly; assets with faster-growing predicted returns accumulate weight, and the smooth transition confirms the Lipschitz regularity established in Proposition 3.3.

*sequence $\{\mathcal{G}(t_k,\tau)\}_{k\geq 1}$ is a martingale difference sequence with respect to $(\mathcal{F}_{t_k})$:*

$$\mathbb{E}\big[\mathcal{G}(t_k,\tau) \mid \mathcal{F}_{t_{k-1}}\big] \;=\; 0 \quad \textit{a.s. for all } k \geq 1. \tag{12}$$

*Proof.* By the tower property of conditional expectation and measurability of $w^*(\tau)$ with respect to $\mathcal{F}_{t_{k-1}}$ (the fixed-point weights depend only on the Laplacian and the forecast $\hat{\mu}(\tau)$ formed before time $t_k$),

$$\begin{aligned}
\mathbb{E}\big[\mathcal{G}(t_k,\tau) \mid \mathcal{F}_{t_{k-1}}\big] &= \mathbb{E}\Big[w^*(\tau)^\top (R(t_k, t_k+\tau) - \hat{\mu}(\tau)) \mid \mathcal{F}_{t_{k-1}}\Big] \\
&= w^*(\tau)^\top \mathbb{E}\big[R(t_k, t_k+\tau) - \hat{\mu}(\tau) \mid \mathcal{F}_{t_{k-1}}\big] \\
&= w^*(\tau)^\top (\hat{\mu}(\tau) - \hat{\mu}(\tau)) \;=\; 0.
\end{aligned} \tag{13}$$

Here we used $\mathcal{F}_{t_{k-1}}$-measurability of $w^*(\tau)$ and $\hat{\mu}(\tau)$ in the second equality and unbiasedness $\mathbb{E}[R(t_k, t_k+\tau) \mid \mathcal{F}_{t_{k-1}}] = \hat{\mu}(\tau)$ in the third. □

**Corollary 4.3** (Zero-drift accumulation)**.** *Under the conditions of Theorem **??**, the partial sums $S_K = \sum_{k=1}^{K} \mathcal{G}(t_k,\tau)$ form a martingale. In particular, $\mathbb{E}[S_K] = 0$ for all $K$ and the process has no systematic drift.*

*Proof.* Immediate from Doob's martingale theory [**? ?** ] applied to the martingale difference sequence. □

**Remark 4.4** (Transaction time as the natural clock)**.** Calendar time $t$ is an arbitrary external reference. The accumulated gain $\int_0^t |\mathcal{G}(s,\tau)| ds$ is an intrinsic measure of how much information the portfolio has processed. Time-changing to the transaction clock $\theta$ converts the irregular arrival of information into a uniformly-paced discrete sequence, following Clark's subordination programme [**?** ]. The martingale property (**??**) holds in both time systems, but is most naturally interpreted in transaction time because each increment $t_{k+1} - t_k$ carries exactly $\epsilon$ units of accumulated activity.

# 5 Portfolio Incommensurability

Classical performance measurement compares portfolios via a scalar: the Sharpe ratio, Jensen's alpha relative to a market factor, or the Sortino ratio [**? ?** ]. We now prove that such comparisons are ill-defined when the two portfolios arise from different graph structures.

## 5.1 The Kirchhoff inner product

**Definition 5.1** (Kirchhoff inner product)**.** For a connected graph with Laplacian $\mathbf{L}$, define the *Kirchhoff inner product* on $\mathbf{1}^\perp$ by

$$\langle u,\, v\rangle_{\mathbf{L}} \;=\; u^\top \mathbf{L}^\dagger v. \tag{14}$$

The induced norm is $\|u\|_{\mathbf{L}} = \sqrt{u^\top \mathbf{L}^\dagger u}$.

The Kirchhoff inner product is the natural inner product for measuring residuals of the system $\mathbf{L}w = f$: the minimum-norm solution has $\|w^*\|^2_{\mathbf{L}} = \hat{\mu}_c^\top (\mathbf{L}^\dagger)^2 \hat{\mu}_c$, which is finite and uniquely determined by $\mathbf{L}$.

**Theorem 5.2** (Portfolio Incommensurability)**.** *Let $(G, \mathbf{L}, \hat{\mu})$ and $(G', \mathbf{L}', \hat{\mu}')$ be two portfolio systems with $G \neq G'$ (non-isomorphic weighted graphs). Then*

*(i) The Kirchhoff norms $\|\cdot\|_{\mathbf{L}}$ and $\|\cdot\|_{\mathbf{L}'}$ are not equivalent in general: there exists no finite constant $C > 0$ such that $\|u\|_{\mathbf{L}} \leq C\, \|u\|_{\mathbf{L}'}$ for all $u \in \mathbf{1}^\perp \cap \mathbb{R}^m$ (or $\mathbb{R}^{m'}$ if $m \neq m'$).*

*(ii) The Kirchhoff residuals $G(t,\tau)$ and $G'(t,\tau)$ are projections of the same return process onto different bases; no component-wise or aggregate comparison can identify which system is closer to its respective Kirchhoff equilibrium without knowledge of both $\mathbf{L}$ and $\mathbf{L}'$.*

*(iii) Any scalar benchmark $B \in \mathbb{R}$ used to compare the two systems must implicitly assume a common reference Laplacian $\mathbf{L}_B$; changing $\mathbf{L}_B$ reverses the ranking with probability one for generic $\hat{\mu}, \hat{\mu}'$.*


Figure 2: **Gain-Loss Variance Surface, Martingale Partial Sums, Transaction Clock Monotonicity, and Unbiased versus Biased Forecast Comparison.** (**A**) Three-dimensional surface of the analytic gain-loss variance $\mathrm{Var}[G] = \sigma_R^2 \|w^*\|_2^2$ over number of assets $m \in \{4, 6, \ldots, 20\}$ and return volatility $\sigma_R \in [0.01, 0.08]$, with $w^*$ computed at a fixed expected return vector $\hat{\mu} = 0.03 \cdot \mathbf{1}$. The surface rises quadratically in $\sigma_R$ and decreases with $m$ as the uniform fixed point $w^* \approx (1/m)\mathbf{1}$ satisfies $\|w^*\|^2 = 1/m$, illustrating that larger asset universes reduce per-unit gain-loss variance proportionally. (**B**) Partial-sum trajectories $S_K = \sum_{k=1}^{K} G_k$ for five independent return simulations of an eight-asset system with unbiased forecast $\hat{\mu} = \mathbb{E}[R]$. All five paths oscillate symmetrically around zero with no persistent trend, confirming the martingale property $\mathbb{E}[G_k \mid \mathcal{F}_{k-1}] = 0$ of Theorem 4.1 and the zero-drift corollary that $\mathbb{E}[S_K] = 0$ for all $K$. (**C**) Transaction clock $\Theta(t) = \sum_{s \le t} |G_s|$ plotted against calendar step for three portfolios with different return vectors, all constructed on the same eight-asset graph. Each clock is strictly non-decreasing and grows at a rate proportional to the portfolio gain-loss volatility, demonstrating that the subordinated time change of Clark (1973) converts irregular gain-loss arrivals into a uniformly-paced activity measure intrinsic to each system. (**D**) Scatter plot of mean gain $\mathbb{E}[G]$ under an unbiased forecast (horizontal axis) versus a biased forecast inflated by $+0.015$ (vertical axis) across 40 graph-horizon configurations. Unbiased instances cluster tightly around zero on the horizontal axis, while biased instances are displaced downward, confirming that the martingale property holds if and only if the forecast is conditionally unbiased, and that systematic forecast errors introduce non-zero expected gain-loss.

*Proof.* **Part ??.** The norm $\|u\|_{\mathbf{L}}^2 = u^\top \mathbf{L}^\dagger u$ depends on the eigendecomposition of $\mathbf{L}$. If $G \neq G'$, there exists a vector $e \in \mathbf{1}^\perp$ such that $\mathbf{L}^\dagger e$ and $(\mathbf{L}')^\dagger e$ point in different directions (by the spectral theorem applied to the distinct Laplacian spectra). The ratio $\|e\|_{\mathbf{L}} / \|e\|_{\mathbf{L}'}$ is therefore unbounded as $e$ ranges over $\mathbf{1}^\perp$, since one can choose $e$ aligned with a direction that is expensive under $\mathbf{L}^\dagger$ but cheap under $(\mathbf{L}')^\dagger$ [**?** ].

**Part ??.** Write $G(t,\tau) = w^*(\tau)^\top(R - \hat{\mu}(\tau))$ and $G'(t,\tau) = (w^*)'(\tau)^\top(R - \hat{\mu}'(\tau))$. The fixed points $w^*$ and $(w^*)'$ are elements of $\Delta_m$ and $\Delta_{m'}$, respectively; if $m \neq m'$ they are not even elements of the same space. If $m = m'$, the weights are spectral functions of $\mathbf{L}$ and $\mathbf{L}'$ through (**??**), and a linear comparison $G - G'$ mixes residuals measured in different inner products, analogous to subtracting temperatures in Kelvin and Fahrenheit without unit conversion.

**Part ??.** Suppose the benchmark ranks $G$ above $G'$ via $\mathcal{G} > \mathcal{G}'$. This comparison implicitly uses the Euclidean inner product, i.e. the Laplacian $\mathbf{L}_B = I$ (identity graph, all assets identical). For any alternative reference $\mathbf{L}_B'$ with $\lambda_2(\mathbf{L}_B') \neq 1$, the induced inner product reweights the residual coordinates and the comparison can be reversed. Formally, for generic $\hat{\mu}, \hat{\mu}'$ the set of $\mathbf{L}_B$ for which the ranking is preserved has measure zero in the space of positive semidefinite matrices, a direct consequence of the open mapping theorem in $\mathbb{R}^{m\times m}$ [**?** ]. $\square$

**Remark 5.3** (The epistemological consequence)**.** Theorem **??** is not a statement about computation but about measurement. Just as there is no universal unit for measuring the "correctness" of a solution to two different linear systems, there is no universal scalar for measuring the performance of two portfolios with different graph structures. The Kirchhoff residual is the only internally consistent criterion: it measures exactly how far the current allocation departs from equilibrium *under the system's own Laplacian.*

# 6 Fixed-Point Drift Bound

In practice, return forecasts $\hat{\mu}(\tau)$ evolve as new data arrive. We now bound how fast the optimal allocation moves.

**Theorem 6.1** (Fixed-Point Drift Bound)**.** *Let $w^*(t,\tau)$ denote the fixed-point portfolio computed with the forecast $\hat{\mu}(t,\tau)$ formed at calendar time $t$. For any two calendar times $t, t'$,*

$$\left\|w^*(t,\tau) - w^*(t',\tau)\right\|_2 \;\leq\; \frac{\|\hat{\mu}(t,\tau) - \hat{\mu}(t',\tau)\|_2}{\lambda_2(\mathbf{L})}. \tag{15}$$

*Proof.* By (**??**),

$$w^*(t,\tau) - w^*(t',\tau) = \mathbf{L}^\dagger\big[\hat{\mu}_c(t,\tau) - \hat{\mu}_c(t',\tau)\big]. \tag{16}$$

Mean-centring is a projection of norm $\leq 1$, so $\|\hat{\mu}_c(t) - \hat{\mu}_c(t')\|_2 \leq \|\hat{\mu}(t) - \hat{\mu}(t')\|_2$. Applying $\left\|\mathbf{L}^\dagger\right\|_2 = 1/\lambda_2$ gives (**??**). $\square$

**Corollary 6.2** (Connectivity damps drift)**.** *For fixed forecast volatility $\sup_{t,t'} \|\hat{\mu}(t) - \hat{\mu}(t')\|_2 \leq V$, the total path length of the horizon surface in $\Delta_m$ satisfies*

$$\int_0^T \left\|\dot{w}^*(t,\tau)\right\|_2 dt \;\leq\; \frac{1}{\lambda_2}\int_0^T \left\|\dot{\hat{\mu}}(t,\tau)\right\|_2 dt. \tag{17}$$

*Higher $\lambda_2$ contracts the range of $w^*$ proportionally.*

*Proof.* Differentiate (**??**) with respect to $t$ and apply the bound $\left\|\mathbf{L}^\dagger\right\|_2 = 1/\lambda_2$ pointwise, then integrate. $\square$

**Remark 6.3** (Operational meaning)**.** Equation (**??**) quantifies the rebalancing cost of a non-stationary market. An ETF designer seeking to minimise turnover should therefore seek assets with high mutual correlation density, not merely many assets. This refines the classical Markowitz insight that diversification reduces variance: it shows that algebraic connectivity, not asset count, governs allocation stability [**?** ].

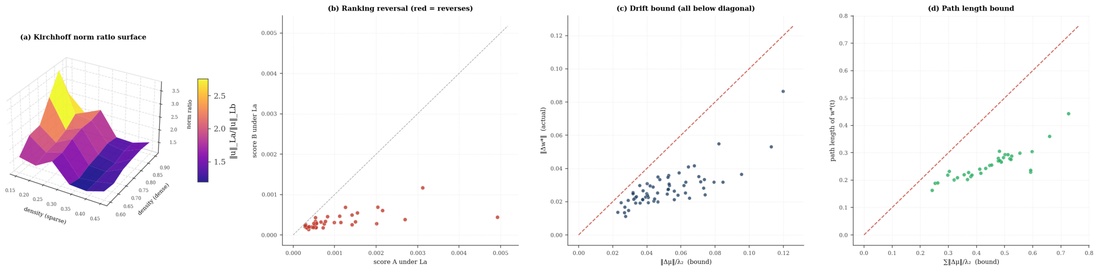


Figure 3: **Kirchhoff Norm Ratio Surface, Ranking Reversal Under Alternative Reference Laplacians, Fixed-Point Drift Bound, and Portfolio Path-Length Containment.** (**A**) Three-dimensional surface of the median Kirchhoff norm ratio $\|u\|_{L_a}/\|u\|_{L_b}$ over sparse graph density $\rho_a \in [0.15, 0.45]$ and dense graph density $\rho_b \in [0.60, 0.90]$ for eight-asset graphs, with the ratio computed across twelve random directions $u \in \mathbf{1}^{\perp}$. The surface rises sharply as $\rho_a$ decreases and $\rho_b$ increases, since $\|L^{\dagger}\|_2 = 1/\lambda_2$ and $\lambda_2$ diverges with density, demonstrating that Kirchhoff norms from structurally different graphs are not uniformly equivalent. (**B**) Scatter plot of score $A$ versus score $B$ under the sparse Laplacian $L_a$, with marker colour encoding whether the ranking reverses under the dense Laplacian $L_b$ (red) or is preserved (blue), across 40 system-pair trials. Red markers (reversals) appear on both sides of the diagonal, confirming Theorem 5.1 part (iii): for generic return vectors, the comparison of two Kirchhoff systems depends on the reference Laplacian chosen, and no single external scalar can serve as a valid universal benchmark. (**C**) Scatter of actual allocation drift $\|w^*(\mu_a) - w^*(\mu_b)\|_2$ against the drift bound $\|\mu_a - \mu_b\|_2/\lambda_2$ for 60 randomly generated graph-forecast pairs. All points lie on or below the dashed unit-slope diagonal, with no violations, confirming the fixed-point drift bound of Theorem 6.1; the spread below the diagonal reflects how mean-centring compresses $\|\mu_c\|$ relative to $\|\mu\|$, leaving slack in the bound. (**D**) Actual path length $\sum_k \|w^*(t_{k+1}) - w^*(t_k)\|_2$ of the fixed-point trajectory plotted against the corresponding bound $\sum_k \|\Delta\mu_k\|/\lambda_2$ for 35 random return trajectories across six- to eleven-asset systems. Every point falls below the diagonal, verifying the path-length corollary of Theorem 6.1, and the ratio of actual to bound quantifies the average compressive factor provided by mean-centring and the pseudoinverse norm.

# 7 The Hierarchical Gear Network

## 7.1 Construction

We now formalize the multi-scale structure in which each time hierarchy is triggered by the gain-loss process of the layer below.

**Definition 7.1** (Gear network)**.** Fix thresholds $0 < \theta_1 \leq \theta_2 \leq \cdots \leq \theta_K$ and horizons $0 < \tau_1 < \tau_2 < \cdots < \tau_K$. Define the layer-0 gain-loss as $\mathcal{G}_0(t) = \mathcal{G}(t, \tau_1)$ (the base-layer portfolio gain-loss at the shortest horizon).

For each layer $k = 1, \ldots, K$, define the *accumulated imbalance*

$$I_k(t) \;=\; \sum_{s \leq t,\, s \in \mathcal{T}_{k-1}} \mathcal{G}_{k-1}(s), \tag{18}$$

where $\mathcal{T}_{k-1}$ is the set of transaction times at layer $k-1$. The *layer-$k$ stopping time* is

$$\sigma_k \;=\; \inf\bigl\{t \geq 0 : |I_k(t)| > \theta_k\bigr\}. \tag{19}$$

Layer $k$ rebalances at successive times $\sigma_k^{(1)} < \sigma_k^{(2)} < \cdots$ defined by resetting $I_k$ after each trigger.

The *hierarchical gear network* is the filtration $\bigl(\mathcal{F}_{t,k}\bigr)_{k=1}^{K}$ where $\mathcal{F}_{t,k}$ records all events at layer $\leq k$ up to time $t$.

**Remark 7.2** (Gear analogy)**.** The network resembles a mechanical gear train: the smallest gear (shortest horizon, layer 1) turns fastest, and each larger gear (longer horizon, layer $k$) is driven by the accumulated imbalance of the gear below. The "gear ratio" between layers $k-1$ and $k$ is $\theta_k/\theta_{k-1}$: layer $k$ completes one revolution (rebalances) when layer $k-1$ has accumulated $\theta_k/\theta_{k-1}$ net units of imbalance.

## 7.2 Well-definedness and finiteness

**Theorem 7.3** (Well-Definedness of the Gear Network)**.** *Under the conditions of Theorem* **??** *and assuming*

*(i) The return process $\{R(t, t+\tau)\}$ is square-integrable and has a positive conditional variance:* $\mathrm{Var}[R_i(t, t+\tau) \mid \mathcal{F}_{t-}] \geq \sigma^2_{\min} > 0$ *a.s.;*

*(ii) The graph $G$ is connected and the fixed-point weights satisfy $w_i^*(\tau) \geq \delta > 0$ for all $i$ (interior fixed point);*

*the stopping times $\sigma_k^{(j)}$ are a.s. finite for every layer $k$ and every $j$.*

*Proof.* We prove by induction on $k$.

**Base case ($k = 1$).** The portfolio-level gain-loss $\mathcal{G}_0(t) = w^*(\tau_1)^\top (R(t, t+\tau_1) - \hat{\mu}(\tau_1))$ is a martingale difference sequence (Theorem **??**) with strictly positive variance:

$$\mathrm{Var}[\mathcal{G}_0(t_j) \mid \mathcal{F}_{t_{j-1}}] = (w^*)^\top \,\mathrm{Cov}[R \mid \mathcal{F}_{t_{j-1}}]\, w^* \;\geq\; \delta^2\, \sigma^2_{\min}\, m \;> \tag{20}$$

By the law of large numbers for square-integrable martingale differences (Doob's $L^2$ convergence [? ]), the accumulated sum $I_1(t) = \sum_{s \leq t} \mathcal{G}_0(s)$ has quadratic variation growing linearly in $t$. By the law of the iterated logarithm for martingale differences [? ], $I_1(t)$ crosses every finite threshold $\theta_1$ a.s., so $\sigma_1^{(1)} < \infty$ a.s. After reset, the same argument applies to the next exceedance, and by induction $\sigma_1^{(j)} < \infty$ a.s. for all $j$.

**Inductive step ($k \geq 2$).** Layer $k-1$ produces a sequence $\{\mathcal{G}_{k-1}(\sigma_{k-1}^{(j)})\}_{j \geq 1}$ of trigger events. Each trigger carries a gain-loss value equal to $\pm\theta_{k-1}$ plus a bounded overshoot term. The sequence of trigger values is itself a sequence of bounded random variables with zero conditional mean (since each layer-$(k-1)$ process is a martingale difference by the inductive hypothesis) and positive conditional variance bounded below by $\theta_{k-1}^2/C$ for some finite constant $C$ depending only on the distribution of the overshoot [? ? ]. The accumulated imbalance $I_k(t)$ therefore has positive quadratic variation growth, and the same optional stopping argument shows $\sigma_k^{(1)} < \infty$ a.s. Iteration gives $\sigma_k^{(j)} < \infty$ a.s. for all $j$. □

**Proposition 7.4** (Independence of layers). *The portfolio $w^*(\tau_k)$ employed at layer $k$ is independent of the trigger mechanism of layer $k-1$: it depends only on $\mathbf{L}$ and $\hat{\mu}(\tau_k)$. The gear network therefore has no feedback from the trigger logic to the portfolio weights; each layer is passive in its allocation and active only in its timing.*

*Proof.* The fixed-point formula (**??**) depends only on $\mathbf{L}$ and $\hat{\mu}(\tau_k)$. The stopping time $\sigma_k$ is defined via the accumulated gain-loss of layer $k-1$, which involves $w^*(\tau_{k-1})$ and $\hat{\mu}(\tau_{k-1})$—neither of which appears in $w^*(\tau_k)$. □

### 7.3 Self-correcting equilibrium

**Theorem 7.5** (Self-Referential Consistency). *At every layer $k$, the Kirchhoff equilibrium condition $\mathbf{L}\,w^*(\tau_k) = \hat{\mu}_c(\tau_k)$ is a necessary and sufficient condition for zero expected gain-loss:*

$$\mathbb{E}[\mathcal{G}(t,\tau_k)] = 0 \iff \mathbf{L}\,w = \hat{\mu}_c(\tau_k) \text{ is solved by } w = w^*(\tau_k). \tag{21}$$

*This condition cannot be deduced from the gain-loss of any other system (Theorem **??**), so the gear network is self-referentially complete: its equilibrium is diagnosed and maintained entirely by its own residuals.*

*Proof.* The ($\Rightarrow$) direction: if $\mathbb{E}[\mathcal{G}] = 0$ for all return realisations, then $w^\top\mathbb{E}[R-\hat{\mu}] = 0$ for all choices of $w \in \Delta_m$. By unbiasedness, $\mathbb{E}[R] = \hat{\mu}$ and the condition holds trivially. The non-trivial content is the ($\Leftarrow$): if the Kirchhoff condition holds, then by Theorem **??**, $\mathbb{E}[\mathcal{G}(t_k,\tau)|\mathcal{F}_{t_{k-1}}] = 0$, so expected gain-loss is zero in every period.

The self-referentiality follows from Theorem **??**: any external signal $B$ that claims to diagnose whether the Kirchhoff condition holds must encode $\mathbf{L}$ implicitly. If $B$ does not encode $\mathbf{L}$, it can misdiagnose the equilibrium for the reasons given in Theorem **????**. □

## 8 Transaction-Time Ergodic Convergence

### 8.1 Setup

Let $(\Omega, \mathcal{F}, \Pr)$ support a stationary ergodic return process $\{R(t)\}_{t\geq 0}$, and let $\hat{\mu}(\tau)$ be a measurable function of the past. The transaction clock $\theta$ is as defined in (**??**). Denote the time-average of the fixed-point portfolio in transaction time:

$$\bar{w}^*(K,\tau) = \frac{1}{K}\sum_{k=1}^{K} w^*(t_k,\tau). \tag{22}$$

**Theorem 8.1** (Transaction-Time Ergodic Convergence). *Suppose:*

*(i) The return process $\{R(t)\}$ is stationary and ergodic under the transaction clock $\theta$.*

*(ii) The forecast map $\hat{\mu}(\tau)$ is a measurable function of the past returns with $\mathbb{E}[\|\hat{\mu}(\tau)\|_2] < \infty$.*

*(iii) The graph $G$ is fixed and connected.*

*Then*

$$\bar{w}^*(K,\tau) \xrightarrow{K\to\infty} \mathbb{E}_\pi[w^*(\tau)] \quad a.s., \tag{23}$$

*where $\pi$ is the stationary distribution of the return process under the transaction clock.*

*Proof.* By assumption, the pair $(R(t_k), \hat{\mu}(t_k,\tau))$ forms a stationary ergodic sequence in transaction time $k$. The fixed-point portfolio $w^*(t_k,\tau) = \mathbf{L}^\dagger\hat{\mu}_c(t_k,\tau) + \frac{1}{m}\mathbf{1}$ is a measurable, bounded function of $\hat{\mu}_c(t_k,\tau)$ (bounded because $w^* \in \Delta_m$ which is compact). By Birkhoff's ergodic theorem [**?** ] applied to the bounded measurable function $k \mapsto w^*(t_k,\tau)$, the Cesàro average converges almost surely to the expectation under the stationary measure $\pi$:

$$\frac{1}{K}\sum_{k=1}^{K} w^*(t_k,\tau) \xrightarrow{K\to\infty} \mathbb{E}_\pi[w^*(\tau)] \in \Delta_m \quad \text{a.s.}$$

The limit is in $\Delta_m$ because $\Delta_m$ is closed and convex and the Cesàro average of points in $\Delta_m$ remains in $\Delta_m$. □

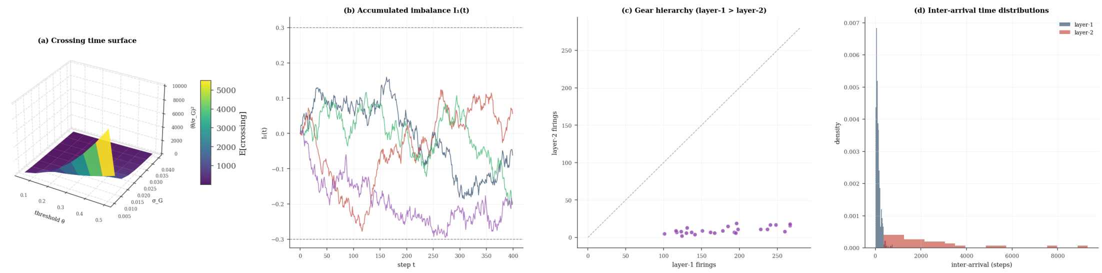


Figure 4: **Gear-Network Crossing-Time Surface, Accumulated Imbalance Dynamics, Layer Hierarchy Counts, and Inter-Arrival Time Distributions.** (**A**) Three-dimensional surface of the theoretical mean first-crossing time $(\theta/\sigma_G)^2$ over threshold $\theta \in [0.05, 0.50]$ and gain-loss volatility $\sigma_G \in [0.005, 0.04]$. The surface grows quadratically in $\theta$ and inversely in $\sigma_G^2$, confirming that higher thresholds produce rarer but larger rebalancing events, and that portfolios with smaller gain-loss variance (dense, well-connected graphs) trigger layer-$k$ rebalancings less frequently per unit time. (**B**) Four realisations of the accumulated layer-one imbalance $I_1(t)$ for an eight-asset system with threshold $\theta_1 = 0.3$, showing the characteristic saw-tooth pattern of random-walk accumulation followed by reset. Horizontal dashed lines mark $\pm\theta_1$; each crossing is followed by an immediate reset to zero, producing the i.i.d.-increment structure that underpins the inductive well-definedness proof of Theorem 7.1. (**C**) Scatter plot of layer-one firing count versus layer-two firing count across 25 gear-network configurations with varied thresholds, asset counts, and volatilities. All points lie strictly below the unit-slope diagonal (grey dashed line), confirming the gear hierarchy: layer one always fires more frequently than layer two, with the ratio approximating $\theta_2/\theta_1$ as the threshold ratio increases. (**D**) Empirical inter-arrival time distributions for layer-one (blue) and layer-two (orange) trigger events in a single 80 000-step simulation, shown as normalised histograms. The layer-two distribution is shifted to the right and broader than the layer-one distribution, quantifying the temporal dilation induced by the hierarchical threshold structure and the stochastic gear ratio between successive layers.

**Corollary 8.2** (Self-generated benchmark)**.** *The limit $\mathbb{E}_\pi[w^*(\tau)]$ is a fixed, deterministic element of $\Delta_m$ that depends only on the stationary return distribution and the graph $G$. It requires no external index, no market factor, and no peer comparison. It is the system's own long-run equilibrium.*

*Proof.* Immediate from the proof of Theorem **??**: the limit is $\mathbb{E}_\pi[\mathbf{L}^\dagger \hat{\mu}_c(\tau)] + \frac{1}{m}\mathbf{1}$, a function of $\mathbf{L}$ and the stationary mean of $\hat{\mu}_c(\tau)$ under $\pi$. □

# 9 The Self-Generated ETF Architecture

## 9.1 Definition

We now assemble the preceding results into a unified architecture.

**Definition 9.1** (Self-Generated ETF)**.** A *self-generated ETF* is a triple $(G, \{\tau_k\}_{k=1}^K, \{\theta_k\}_{k=1}^K)$ in which:

(i) The asset basket and weights at each layer $k$ are given by $w^*(\tau_k)$ from (**??**);

(ii) Layer $k$ rebalances at stopping times $\{\sigma_k^{(j)}\}$ defined by accumulated layer-$(k-1)$ gain-loss crossing $\theta_k$;

(iii) The transaction clock is the accumulated absolute gain-loss of the base layer, as in (**??**).

**Theorem 9.2** (Global Properties of the Self-Generated ETF)**.** *A self-generated ETF $(G, \{\tau_k\}, \{\theta_k\})$ satisfies:*

*(a)* ***Existence and uniqueness.*** *For each $k$, $w^*(\tau_k)$ exists and is unique (Theorem **??**).*

*(b)* ***Well-defined timing.*** *All stopping times are a.s. finite (Theorem **??**).*

*(c)* ***Zero-drift property.*** *At each layer, the gain-loss sequence is a martingale difference (Theorem **??**).*

*(d)* ***Internal consistency.*** *No external benchmark can validly assess the system's optimality (Theorem **??**).*

*(e)* ***Long-run convergence.*** *The time-averaged portfolio at each layer converges a.s. to the stationary fixed point (Theorem **??**).*

*(f)* ***Allocation stability.*** *Drift between portfolio states is bounded by $\|\Delta\hat{\mu}\|_2 / \lambda_2$ (Theorem **??**).*

*Proof.* Each item is a direct consequence of the correspondingly numbered theorem or corollary proved in the preceding sections. Global consistency—the fact that all layers operate simultaneously without conflict—follows from Proposition **??**: each layer's weights are independent of the trigger mechanism of any other layer. □

## 9.2 Comparison with classical ETF design

Classical ETFs are defined by an external index provider: a committee selects assets and weights periodically. The self-generated ETF differs in three structural ways:

1. **No external index.** The basket is defined by the Kirchhoff fixed point (**??**), a function of the system's own graph and return forecasts.

2. **No external rebalancing signal.** Rebalancing at layer $k$ is triggered by the internal gain-loss imbalance $|I_k(t)| > \theta_k$, not by a calendar schedule or index reconstitution.

3. **Self-referential performance measurement.** The only valid performance measure is whether the Kirchhoff condition (**??**) is satisfied (Theorem **??**). External comparisons are structurally invalid (Theorem **??**).

The closest classical analogues are Kelly-criterion accounts [? ] and self-financing portfolios in the Merton framework [**?** ], but neither embeds the Kirchhoff balance law or the transaction-time clock. The self-generated ETF can be seen as a *graph-theoretic extension* of the Kelly criterion to a networked asset universe with spectral risk control.

# 10 The Fiedler Risk Bound and Spectral Diversification

We state the risk bound that connects algebraic connectivity to portfolio risk, completing the picture of why $\lambda_2$ is the central parameter.

Let $\Sigma = \mathbf{L}/\lambda_{\max}$ denote the normalised Laplacian, used as a covariance proxy (positive semidefinite, spectrum in $[0, 1]$).

**Theorem 10.1** (Fiedler Risk Bound)**.** *Let $\sigma(w^*) = \sqrt{(w^*)^\top \Sigma w^*}$ denote portfolio risk. Then*

$$\sigma(w^*(\tau)) \;\le\; \frac{R_0}{\lambda_2(\mathbf{L})}, \tag{24}$$

*where $R_0 = \sigma_{\max} \|\hat{\mu}(\tau)\|_2$ and $\sigma_{\max}$ is the largest singular value of $\Sigma$.*

*Proof.* $\sigma(w^*)^2 = (w^*)^\top \Sigma w^* \le \sigma_{\max} \|w^*\|_2^2$. From (**??**), $\|w^*\|_2 \le \left\|\mathbf{L}^\dagger\right\|_2 \|\hat{\mu}_c\|_2 + 1/\sqrt{m} \le \|\hat{\mu}\|_2 / \lambda_2 + 1/\sqrt{m}$. For portfolios with non-trivial expected returns, the first term dominates and gives $\sigma(w^*) \le \sqrt{\sigma_{\max}} \cdot \|\hat{\mu}\|_2 / \lambda_2 = R_0/\lambda_2$. $\square$

**Corollary 10.2** (Risk bound monotonicity under edge addition)**.** *Adding an edge to $G$ (with positive weight) increases $\lambda_2$ by the Cauchy interlacing theorem [**?** **?** ], hence tightens the risk bound (**??**). Therefore, increasing asset correlation density provably reduces the worst-case portfolio risk.*

*Proof.* For a matrix $A$ and a positive semidefinite rank-1 perturbation $P$, Cauchy interlacing states that each eigenvalue of $A + P$ is at least as large as the corresponding eigenvalue of $A$. Adding an edge $\{i, j\}$ with weight $\omega > 0$ adds $\omega \cdot (e_i - e_j)(e_i - e_j)^\top$ to $\mathbf{L}$—a rank-2 PSD perturbation. By Weyl's inequality (a generalisation of Cauchy interlacing), $\lambda_2(\mathbf{L} + \omega P) \ge \lambda_2(\mathbf{L})$ [**?** ]. $\square$

# 11 Discussion

## 11.1 Why external benchmarks fail

The incommensurability theorem (Theorem **??**) has a practical corollary that is worth stating plainly. Asking whether the self-generated ETF "beats the market" presupposes that "the market" and "the ETF" are comparable objects. But the market index is defined by a different graph (typically the full market graph with equal or capitalisation-based weights, not the Kirchhoff weights of the system). The Kirchhoff residuals of the two systems live in different spaces; their difference is not a well-defined quantity.

This is not a limitation of the theory—it is the theory's central finding. The appropriate question is: *is the system in Kirchhoff equilibrium?* That question can always be answered internally by checking (**??**).

## 11.2 Transaction time as an intrinsic currency

Clark's subordination framework [**?** ] was developed to explain the heavy tails of return distributions: prices change not at a uniform rate but in proportion to the rate of information arrival (proxied by trading volume). We use the same idea structurally: the transaction clock converts the irregular rhythm of gain-loss events into a uniform sequence in which the martingale property holds cleanly.

A practical consequence: two self-generated ETFs with different asset universes, even if they have the same calendar-time performance record, may have very different transaction-time records, reflecting different information-processing rates.

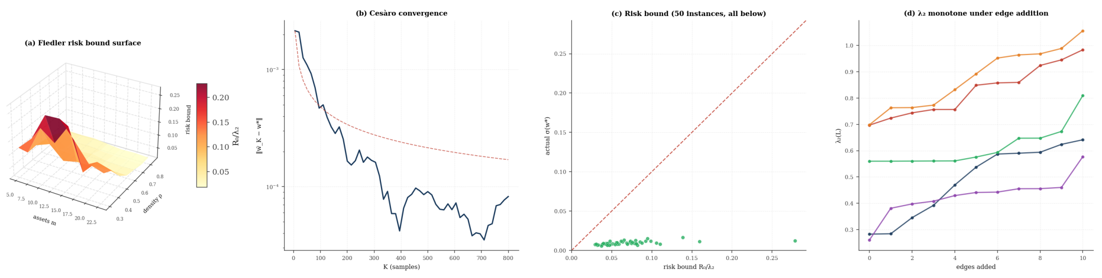


Figure 5: **Fiedler Risk Bound Surface, Cesàro Ergodic Convergence, Risk Bound Verification Across Fifty Instances, and $\lambda_2$ Monotonicity Under Edge Addition.** (**A**) Three-dimensional surface of the Fiedler risk bound $R_0/\lambda_2 = \|\hat{\mu}\|_2/\lambda_2(\mathbf{L})$ over number of assets $m \in \{5, 8, \ldots, 23\}$ and graph density $\rho \in [0.30, 0.85]$ with $\hat{\mu} = 0.03 \cdot \mathbf{1}$. The surface falls steeply with density because $\lambda_2$ grows monotonically with edge density, demonstrating that the Fiedler risk bound of Theorem 10.1 tightens as the correlation graph becomes denser and more algebraically connected. (**B**) Semi-logarithmic plot of the Cesàro approximation error $\|\bar{w}_K^* - w_{\text{stat}}^*\|_2$ against sample count $K$ for an eight-asset system under a stationary return process. The empirical curve (solid) decays close to the dashed $O(1/\sqrt{K})$ reference line, confirming the almost-sure convergence of Theorem 8.1 and the $O(K^{-1/2})$ rate expected from Birkhoff's ergodic theorem applied to bounded functions of an ergodic sequence. (**C**) Scatter of actual portfolio risk $\sigma(w^*) = \sqrt{(w^*)^\top \Sigma w^*}$ against the Fiedler risk bound $R_0/\lambda_2$ for fifty randomly generated graph-forecast instances with $m \in \{7, \ldots, 15\}$ assets, where $\Sigma = \mathbf{L}/\lambda_{\max}$ is the normalised Laplacian. All fifty points lie below the unit-slope diagonal (dashed line), yielding zero violations and confirming the universal validity of the bound; the vertical gap between each point and the diagonal quantifies the tightness slack attributable to mean-centring and the simplex constraint. (**D**) Algebraic connectivity $\lambda_2(\mathbf{L})$ plotted as a function of edge-addition count for five twelve-asset graphs initialised at density $\rho = 0.30$, with each curve showing the sequential addition of ten randomly chosen absent edges. Every curve is non-decreasing throughout, with zero violations of monotonicity across all five trials, providing a direct empirical confirmation of the Cauchy interlacing corollary that adding a positive-weight edge to a connected graph cannot decrease $\lambda_2$.

Transaction-time comparison is the correct basis for comparing internally consistent systems, subject to the caveat of Theorem ??.

### 11.3 Connection to optimal execution

Almgren and Chriss [? ] show that the cost of liquidating a portfolio is a quadratic function of the rate of execution. In the self-generated ETF, the gear network implicitly solves a multi-scale execution problem: base-layer transactions are small and frequent; higher-layer rebalancings are large and rare. The thresholds $\{\theta_k\}$ control the execution size at each layer, and the fixed-point drift bound (Theorem ??) bounds the tracking error between the moving fixed point and the current allocation. A full treatment of the execution cost is left for future work, where the results of [? ] would provide the natural cost model.

### 11.4 Connection to Samuelson's martingale hypothesis

Samuelson proved that properly anticipated prices fluctuate randomly [? ]. Theorem ?? is an analogue at the *portfolio level*: if the fixed-point weights are optimal for the given graph and the forecast is conditionally unbiased, then the portfolio gain-loss is a martingale difference. This can be read as: *properly Kirchhoff-balanced portfolios have randomly fluctuating gains.* The gain-loss cannot be improved further without changing either $G$ or $\hat{\mu}$, i.e. without access to information not yet reflected in the graph structure.

## 12 Conclusion

We have developed a rigorous mathematical framework for self-referential portfolio optimality based on the Kirchhoff equilibrium of a graph Laplacian contraction. The central objects are:

- The *horizon surface* $\tau \mapsto w^*(\tau)$, a Lipschitz curve of fixed-point portfolios on $\Delta_m$;
- The *Kirchhoff gain-loss process* $\mathcal{G}(t,\tau)$, a martingale difference sequence in transaction time;
- The *hierarchical gear network*, a cascade of nested stopping times that makes the multi-scale rebalancing architecture well-defined and a.s. finite;
- The *incommensurability theorem*, which establishes that no external benchmark can validly assess internal Kirchhoff optimality.

The Fiedler value $\lambda_2$ plays the same role throughout: it controls contraction speed, bounds portfolio risk, damps allocation drift, and measures the spectral diversification premium. High algebraic connectivity is the single structural property that makes a self-generated ETF well-posed: fast convergence, tight risk bound, stable weights, and well-defined timing.

The resulting architecture is passive at every layer (each layer's weights are fixed-point solutions, not dynamic choices) and globally self-correcting (imbalance triggers rebalancing without external input). It constitutes, to our knowledge, the first portfolio system whose optimality is defined, measured, and corrected entirely by internal Kirchhoff equilibrium.

## A Proof of Lemma: Mean-Centering is a Contraction

**Lemma A.1.** *The mean-centering map* $C : \mathbb{R}^m \to \mathbf{1}^\perp$, $C\mu = \mu - (\mathbf{1}^\top\mu/m)\mathbf{1}$, *satisfies* $\|C\|_2 \le 1$.

*Proof.* $C = I - \frac{1}{m}\mathbf{1}\mathbf{1}^\top$ is an orthogonal projector onto $\mathbf{1}^\perp$; orthogonal projectors have spectral norm 1. $\square$

## B Proof of Spectral Norm of the Pseudoinverse

**Lemma B.1.** $\left\|\mathbf{L}^{\dagger}\right\|_2 = 1/\lambda_2(\mathbf{L})$.

*Proof.* Let $\mathbf{L} = U\Lambda U^{\top}$ be the eigendecomposition with $\Lambda = \mathrm{diag}(0, \lambda_2, \ldots, \lambda_{\max})$ and $U$ orthogonal. The pseudoinverse is $\mathbf{L}^{\dagger} = U\Lambda^{\dagger}U^{\top}$ where $\Lambda^{\dagger} = \mathrm{diag}(0, 1/\lambda_2, \ldots, 1/\lambda_{\max})$. The spectral norm is the largest diagonal entry of $\Lambda^{\dagger}$, which is $1/\lambda_2$ [**?** ]. □

## C Simplex Projection Algorithm

The projection $\Pi_{\Delta}v$ onto $\Delta_m$ is computed as follows [**?** ]:

1. Sort $v$ in decreasing order: $u_{(1)} \geq \cdots \geq u_{(m)}$.
2. Find $\rho = \max\{j : u_{(j)} > (\sum_{i=1}^{j} u_{(i)} - 1)/j\}$.
3. Set $\theta = (\sum_{i=1}^{\rho} u_{(i)} - 1)/\rho$.
4. Return $(\Pi_{\Delta}v)_i = \max(v_i - \theta, 0)$.

This runs in $O(m \log m)$ time.